Types of Agent :

1. function TABLE-DRIVEN-AGENT (percept) return an action

persistent: percepts, a sequence, initially empty
table, a table of actions, indexed by percept sequences, initially fully specified

append percept to the end of percepts
action ← LOOKUP (percepts, table)

return action.

2. function SIMPLE-REFIEX-AGENT(percept) return an action

persistent: rules, a set of condition-action rules

state ← INTERPRET-INPUT (percept)
rule ← RULE-MATCH (state, rules)
action ← rule. ACTION

return action.

3. function MODEL-BASED-REFLEX-AGENT (percept) returns an action

persistent: state, the agent's current conception of the world state,
model, a description of how the next state depends on current state and action.
rules, a set of condition-action rules.
action, the most recent action, initially none.

state ← UPDATE-STATE (state, action, percept, model)

rule ← RULE-MATCH (state, rules)

action ← rule. ACTION

return action

4. function GOAL-BASED-AGENT (percept) returns an action

persistent: state, the ~~p~~ agent's current conception of the world state.
model, a description of how the next state depends on current state and action
~~goa~~ goals, the agent's set of goals.
plan, the current plan to achieve the goals, initially empty.
action, the most recent action, initially none.

state ← UPDATE-STATE(state, percept, action, model)
if GOAL-ACHIEVED (state, goal) then return a null action
if plan is empty or ~~not~~ plan is not valid in state:

~~goals ← FORMULATE-GOALS (state)~~

plan ← FORMULATE-PLAN (state, goal's) model)
action ← FIRST (plan)

~~action ← EXECUTE-PLAN (plan)~~
plan ← REST (plan)

return action.

5. function UTILITY_BASED_AGENT (percept) returns an action

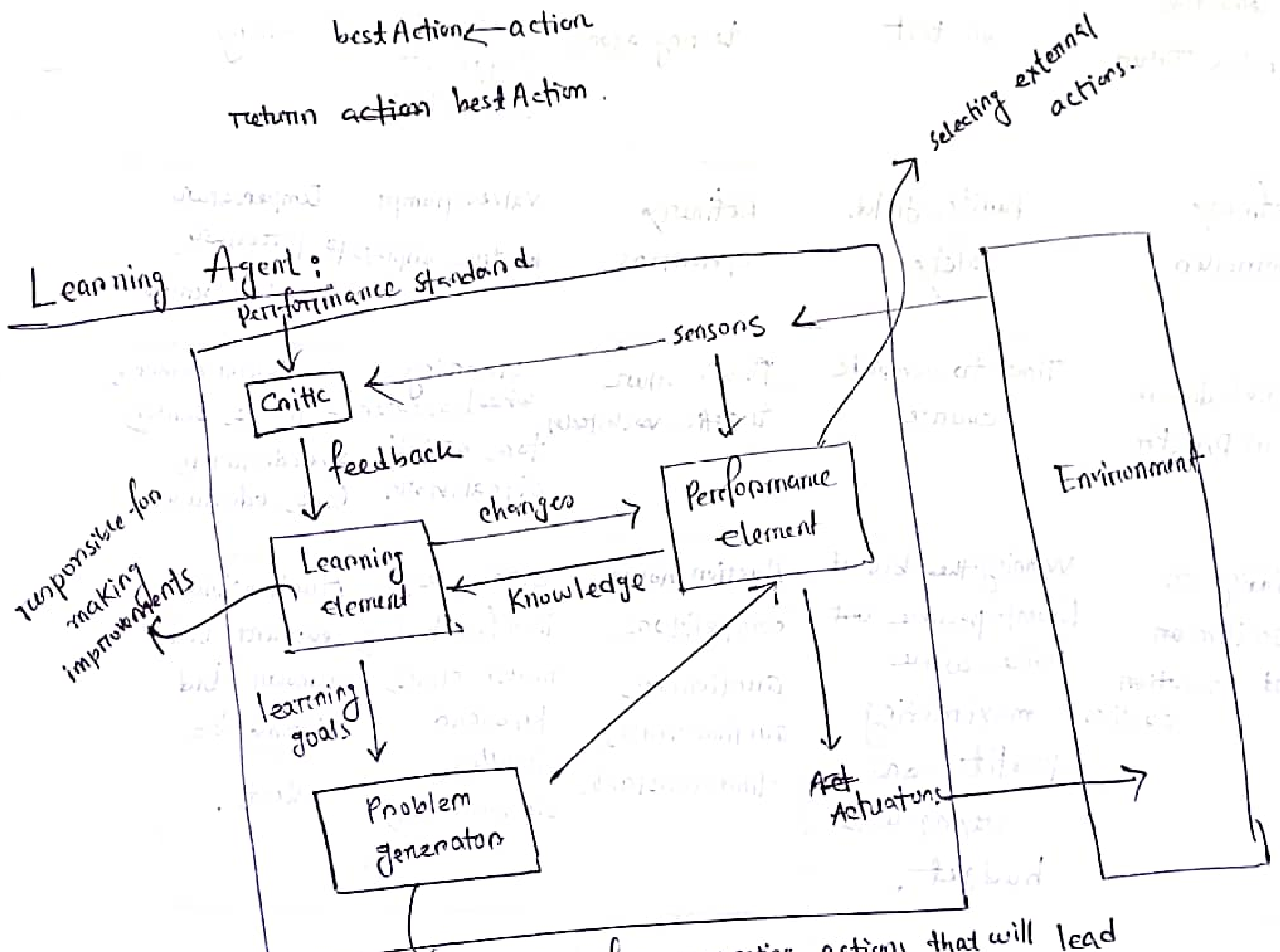
persistent: state, the agent's current conception of the world state

```
state ← UPDATE_STATE (state, percept)
actions ← GET_POSSIBLE_ACTIONS (state)
utility ← - Infinity
bestAction ← NoOP

for action in actions:
    newUtility ← CALCULATE_UTILITY (state, action)
    if newUtility > utility
        utility ← new Utility
        ~~bestA~~
    bestAction ← action

return ~~action~~ bestAction .
```

Learning Agent:



PEAS : (Performance measure, Environment, Actuators, Sensors)

1.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical Diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers.
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	~~Dept~~ Display of scene categorization	color pixel arrays
Part-picking Robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Joined arm and hand	camera, joint angle sensors
Interactive English Tutor	student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections.	Keyboard entry
Refinery controller	Purity, yield, safety	Refinery, operators	valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Robot drivers in DARPA	Time to complete course	Roads, other traffic, ~~w~~obstacles	Steering wheel, accelerator, break, signal, horn	optical cameras, lasers, sonar, speedometer, GPS, odometer
Bidding on an item ~~at~~ at an ~~action~~ auction	Winning the bid at lowest possible ~~cost~~ price while maximizing profit and staying with budget.	Auction house, competitors, auctioneer, auction rules, time constraints.	Bidding system interface (button, mouse clicks, keyboard input), bid increments,	Auction timer, current bid amount, bid history, bid alerts.

* function UTILITY-BASED-AGENT (percept) returns an action

persistent: state, the agent's current conception of the world state

model, a description of how the next state depends on the current state and action.

. utility-function, a description of the agent's utility function.

plan, a sequence of actions to take, initially empty.

action, the most recent action, initially none.

state ← UPDATE-STATE (state, action, percept, model)

if plan is empty then

plan ← PLAN (state, utility-function, model)

action ← FIRST (plan)

plan ← REST (plan)

return action.